

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: Procl-Rev-Ci.sil etc.

Creation date: 18-Mar-80 15:37:56

For: Spruce

35 total sheets = 34 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.

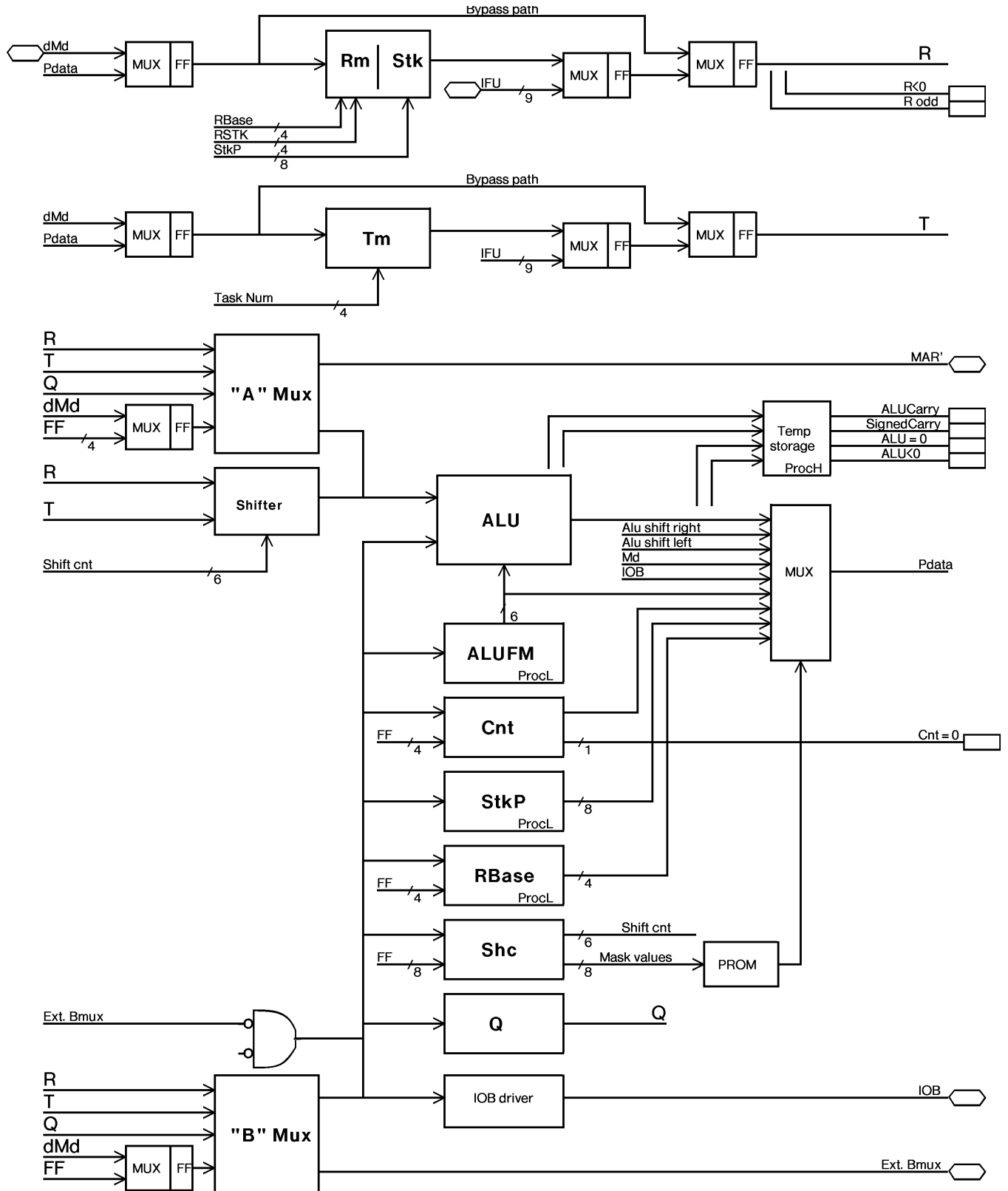
DORADO SCHEMATICS

Low Byte

PROCESSOR

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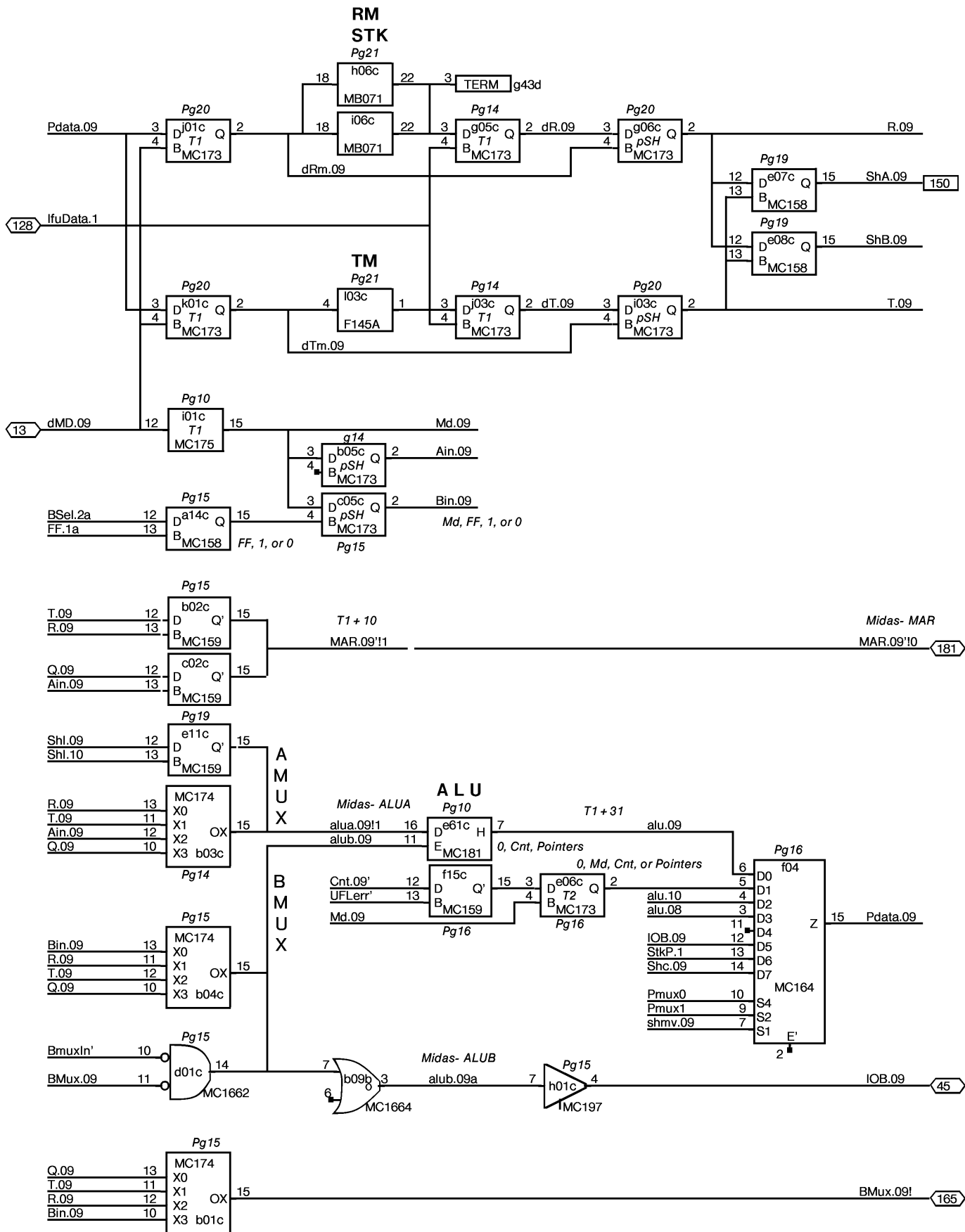
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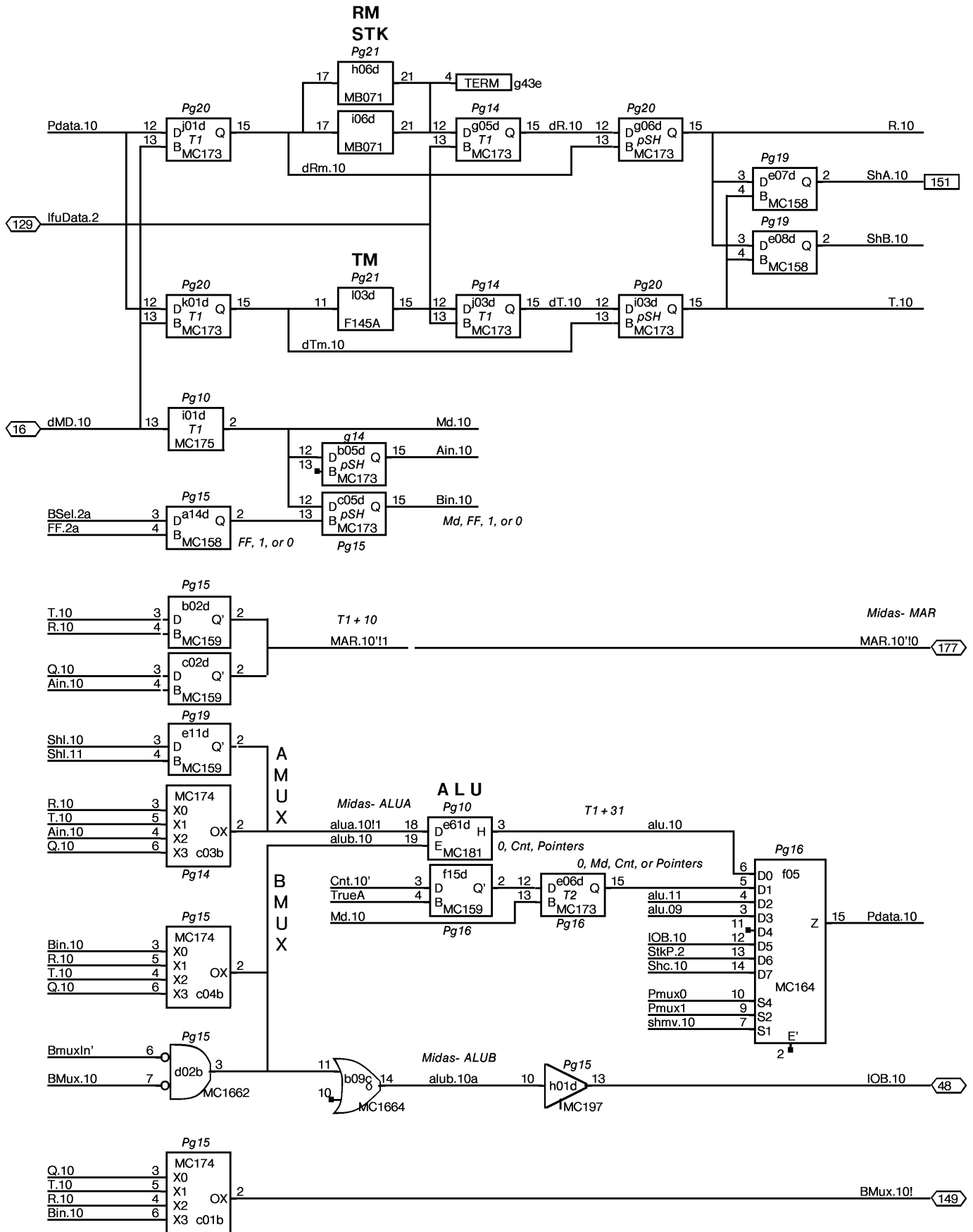


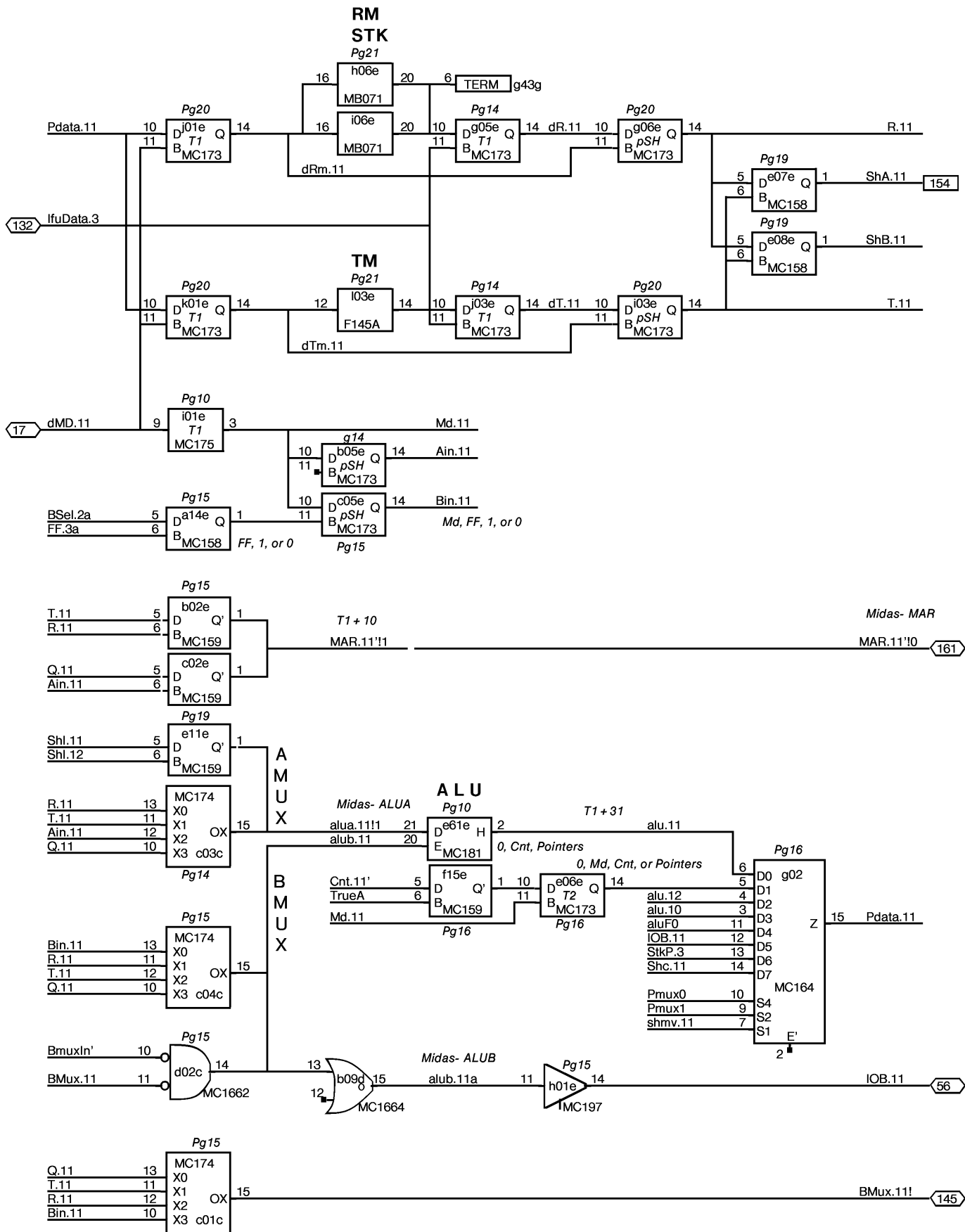
Note: all busses 16 wide unless noted otherwise

XEROX PARC	Project	Reference	File	Designer	Rev	Date	Page
	Dorado	Block Diagram	ProcL01.sil	R Bates	Ci	6/27/79	01















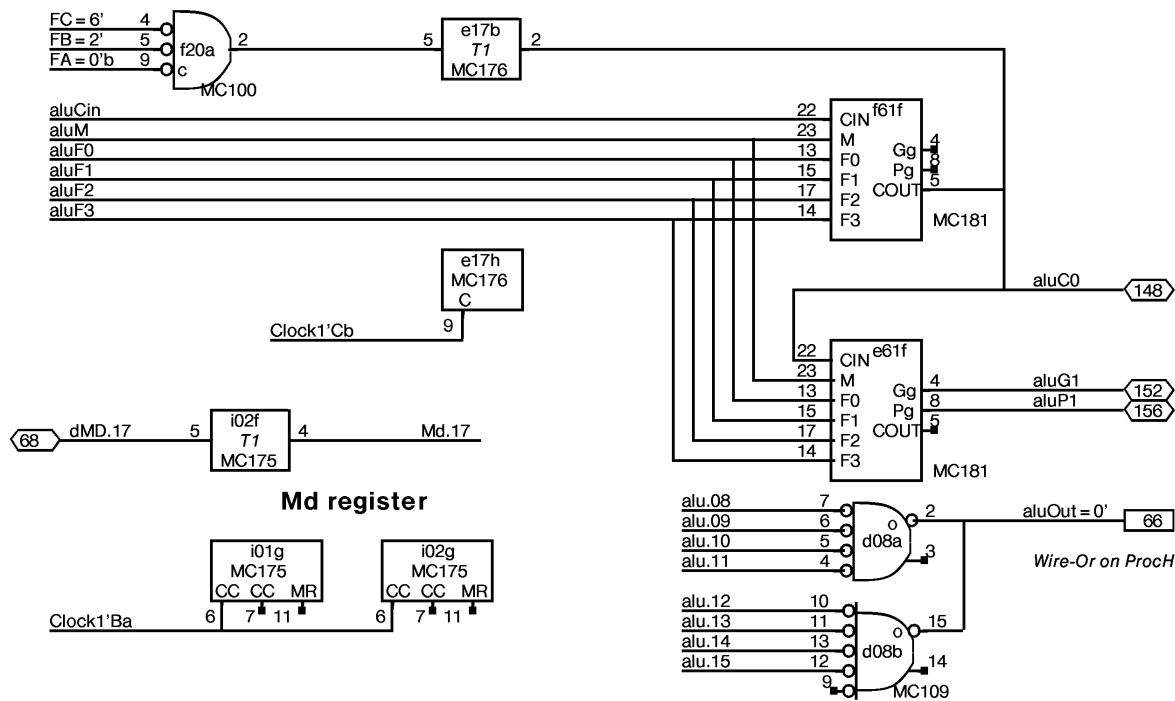


ALU delays

Logical function to output = 11.9
 Arithmetic operation to data = 20.0
 Arithmetic operation to carry = 17.9

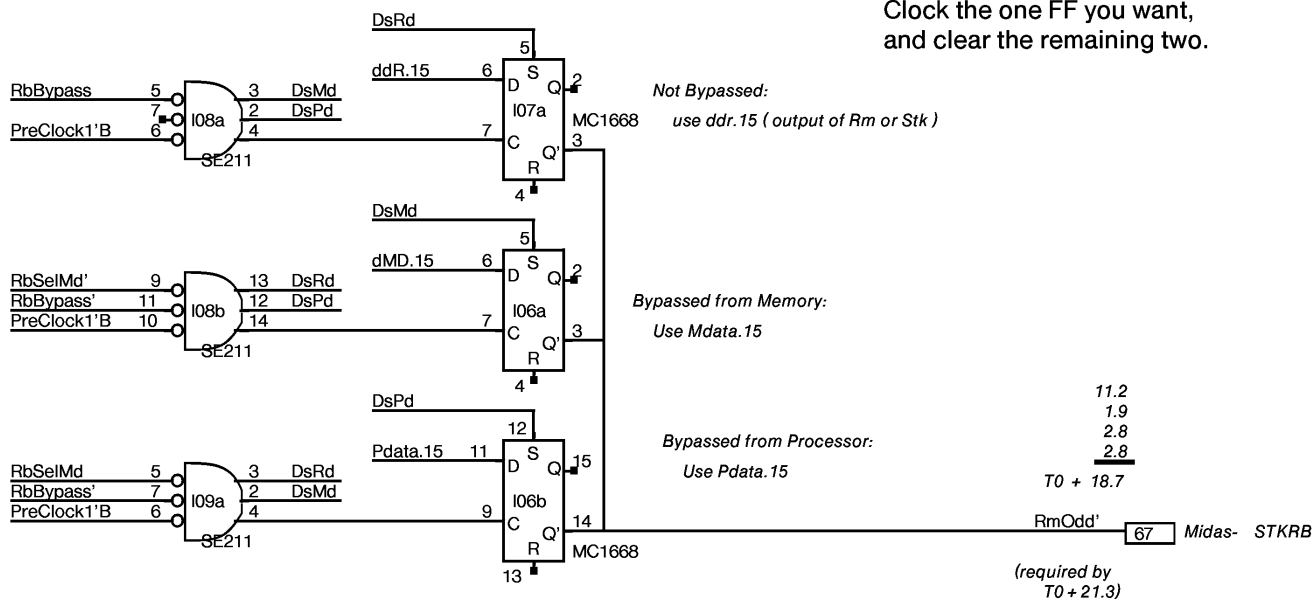
ALU output (assuming 10.2 ns to output of BMux)

Logical Function = 22.0
 Arithmetic operation to data = 30.1
 Arithmetic operation to carry = 28.0



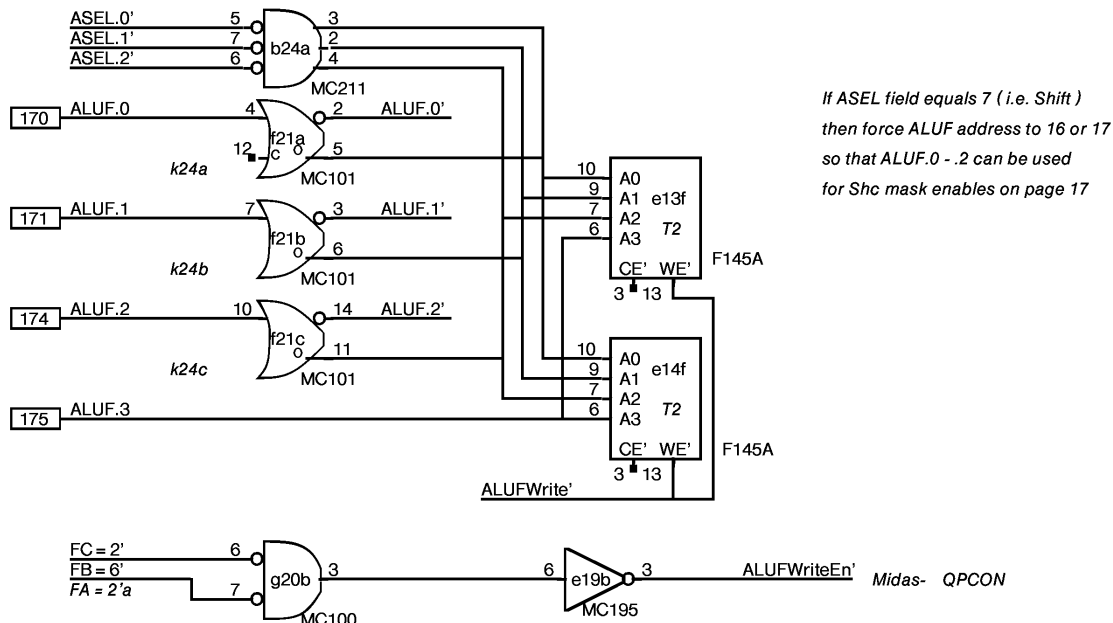
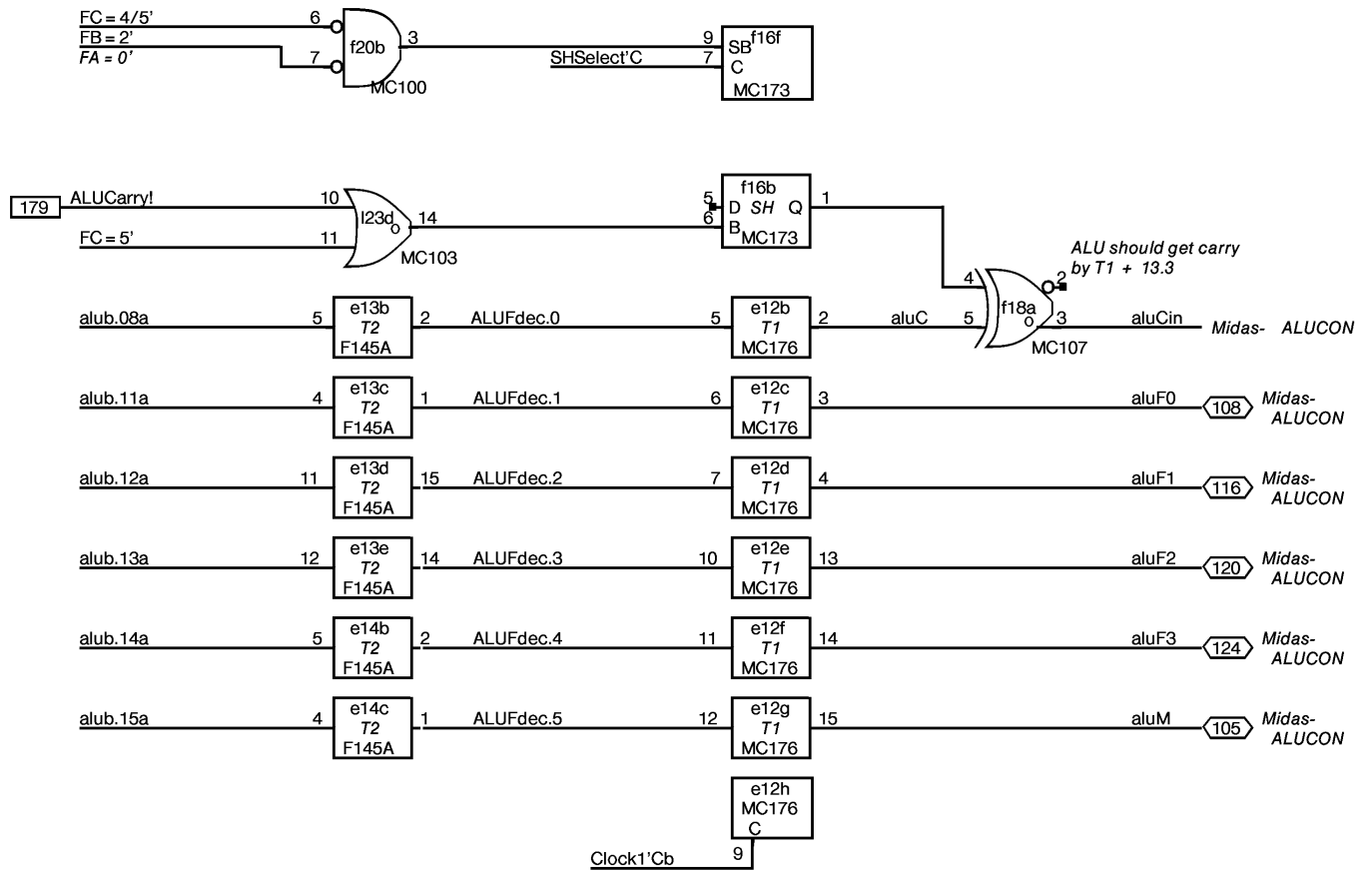
Fast Branch on Rm odd

Clock the one FF you want,
 and clear the remaining two.

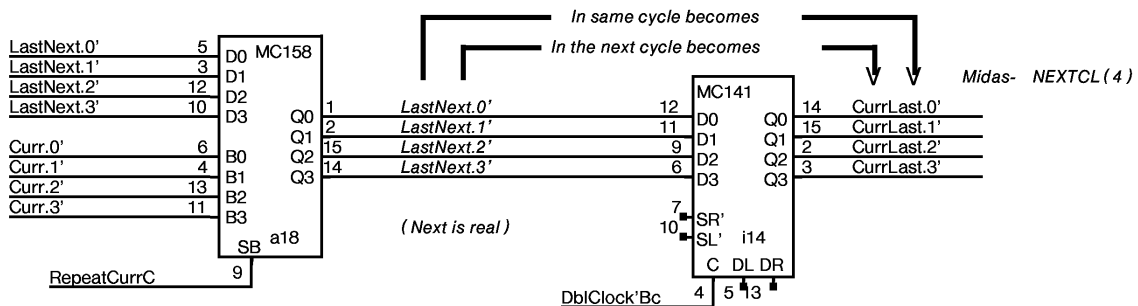
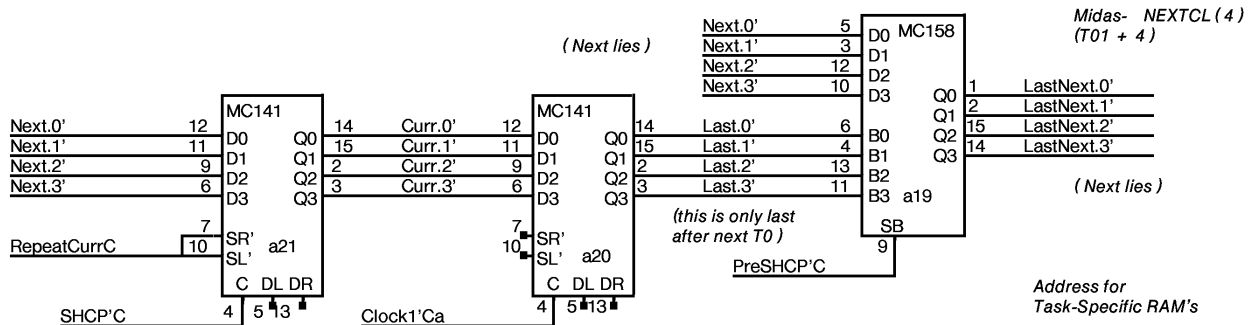
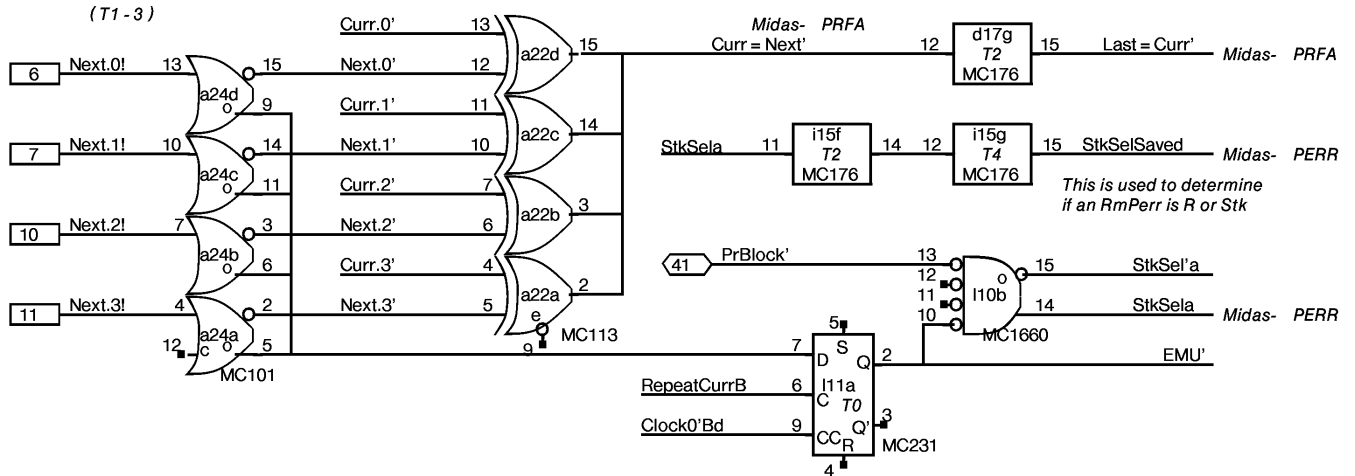


This circuit will correctly
 bypass R from Pdata or Mdata
 When "RisIfData" is in effect, the fast branch
 will be based on the contents of the addressed
 RM or STK, bypassed if necessary.

FF = Cin xor 1
or
FF = Cin xor Cout



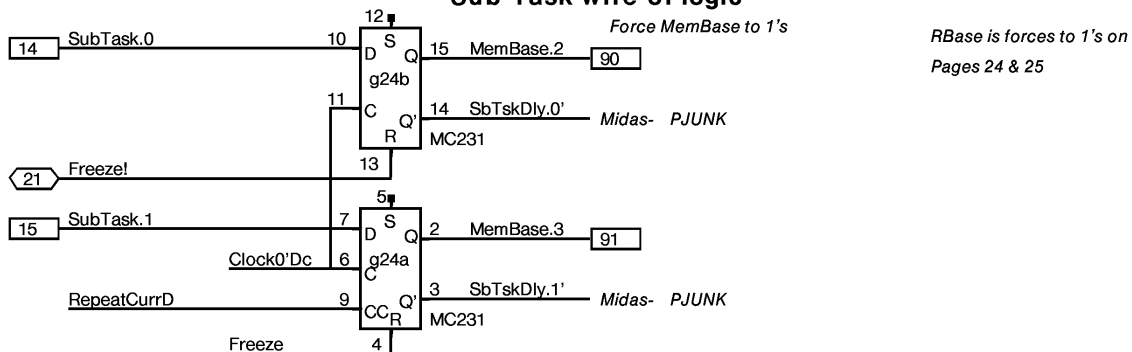
NOTE: Moving k24 to f21 saves 0.4 ns

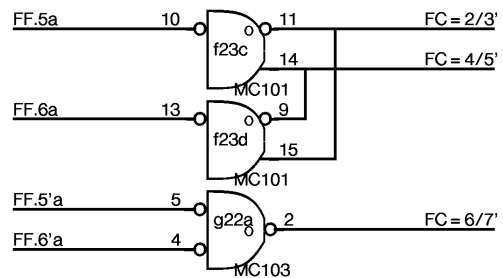
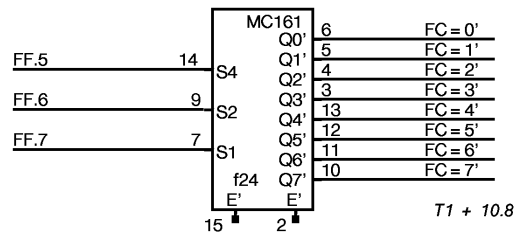
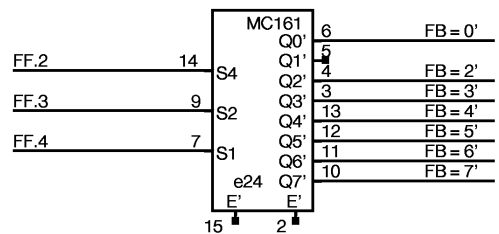
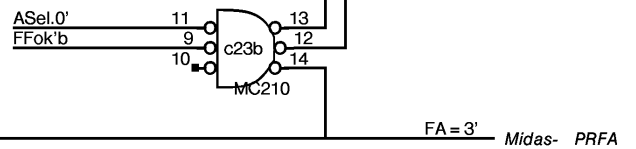
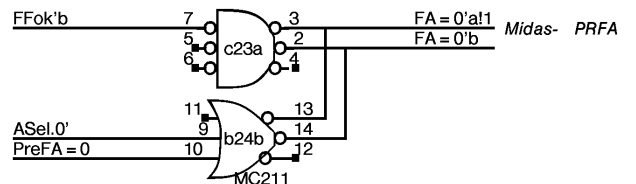
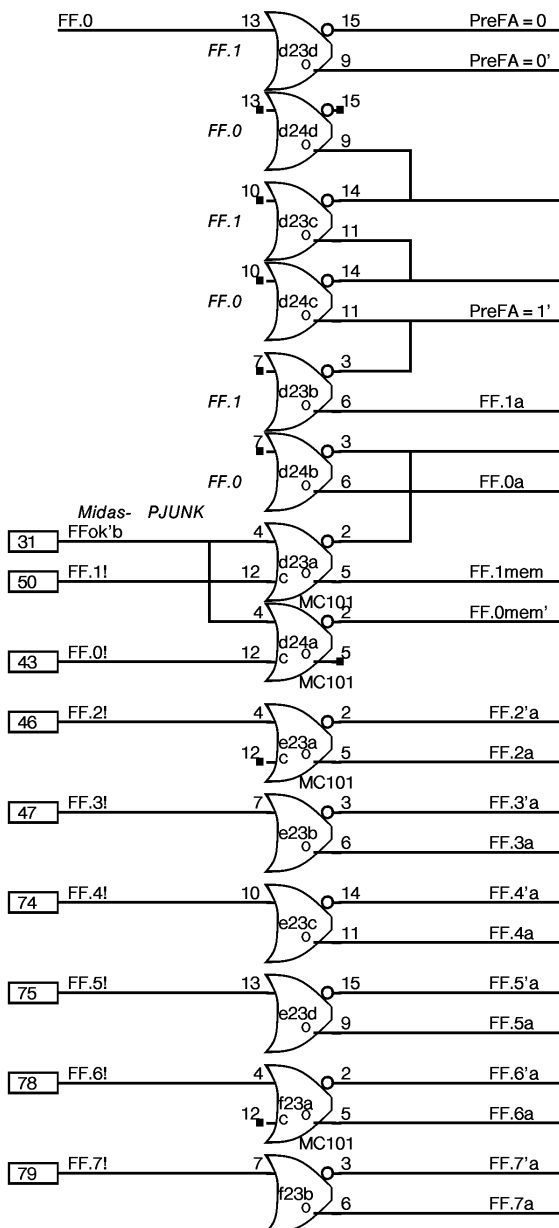
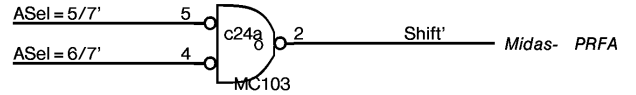
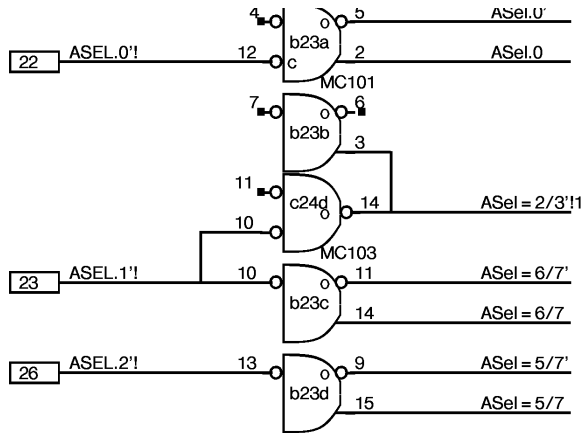


"RepeatCurrent" is asserted on certain occasions when the value on the NEXT bus may be invalid ("Next Lies") due to the combination of Block and Hold.

Task number tracking logic

Sub-Task wire-or logic





AMux decoding

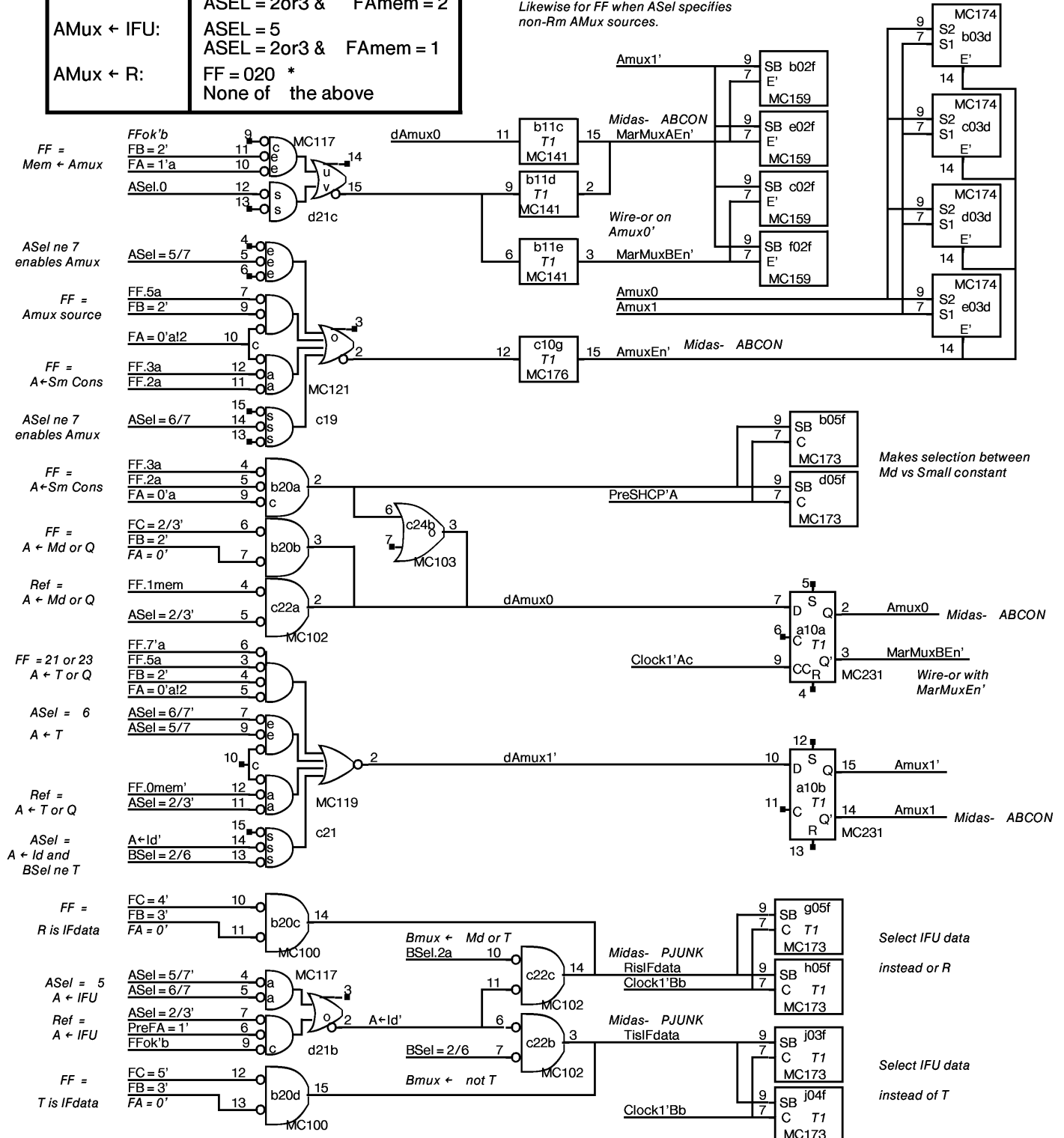
AMux ← FF:	FF = 0-17 *
AMux ← T:	FF = 021 * ASEL = 2or3 & FAmem = 3 ASEL = 6
AMux ← Md:	FF = 022 * ASEL = 2or3 & FAmem = 0
AMux ← Q:	FF = 023 * ASEL = 2or3 & FAmem = 2
AMux ← IFU:	ASEL = 5 ASEL = 2or3 & FAmem = 1
AMux ← R:	FF = 020 * None of the above

* The Amux is disabled by ASEL = 7 unless one of these codes are in effect

NOTE: ASEL selects and FF selects for the AMux are "OR'd" by this hardware. Thus ASEL codes selecting non-Rm sources of Amux must not be used when an FF specifies an ASEL source. Likewise for FF when ASEL specifies non-Rm AMux sources.

AMux encoding

Mux Input	Source
0	R or IFU data
1	T or IFU data
2	Md or Small Const.
3	Q

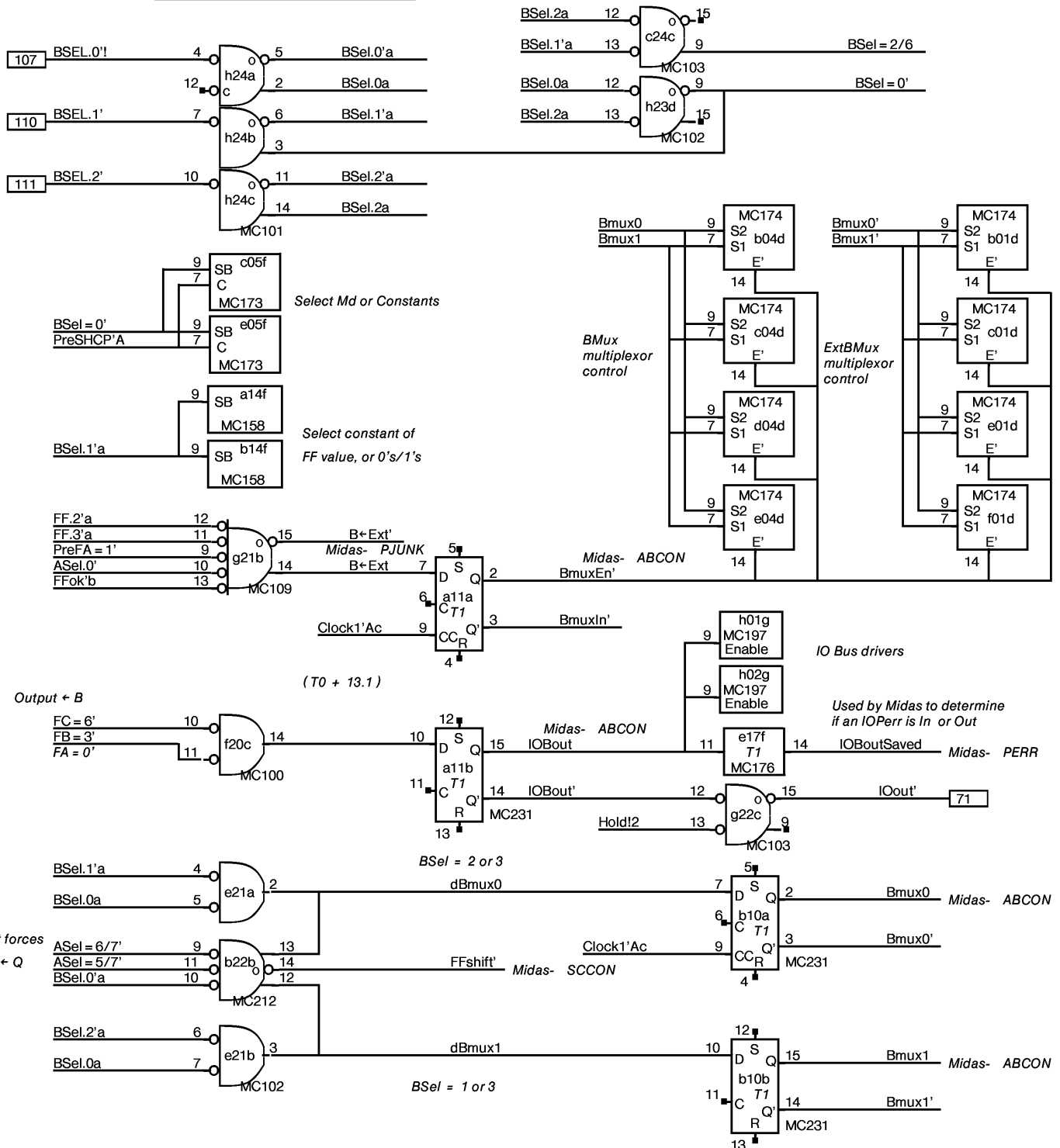


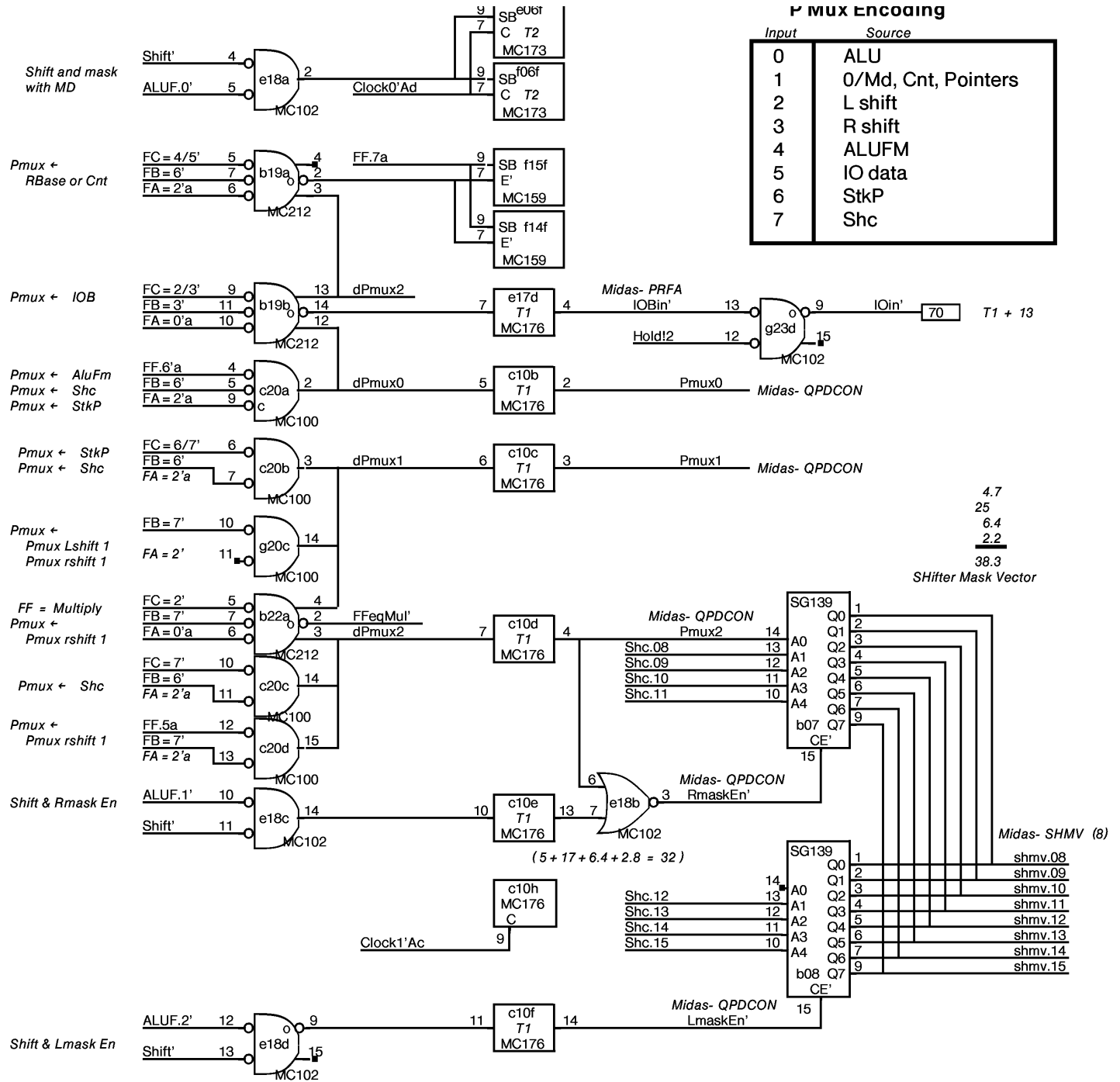
BSEL field decoding

BSEL	INTERNAL	EXTERNAL
0	Md	--
1	R	--
2	T	Hold ← B
3	Q	Q ← B
4	0,,FF	--
5	377,FF	--
6	FF,,0	--
7	FF,,377	--

BMux encoding

BMux	
0	Md or Constant
1	R
2	T
3	Q





NOTE: The prom patterns are designed so that a one into address bit 0 will produce all 1's on the output. This allows the odd address inputs to Pmux to be selected.

Lmask

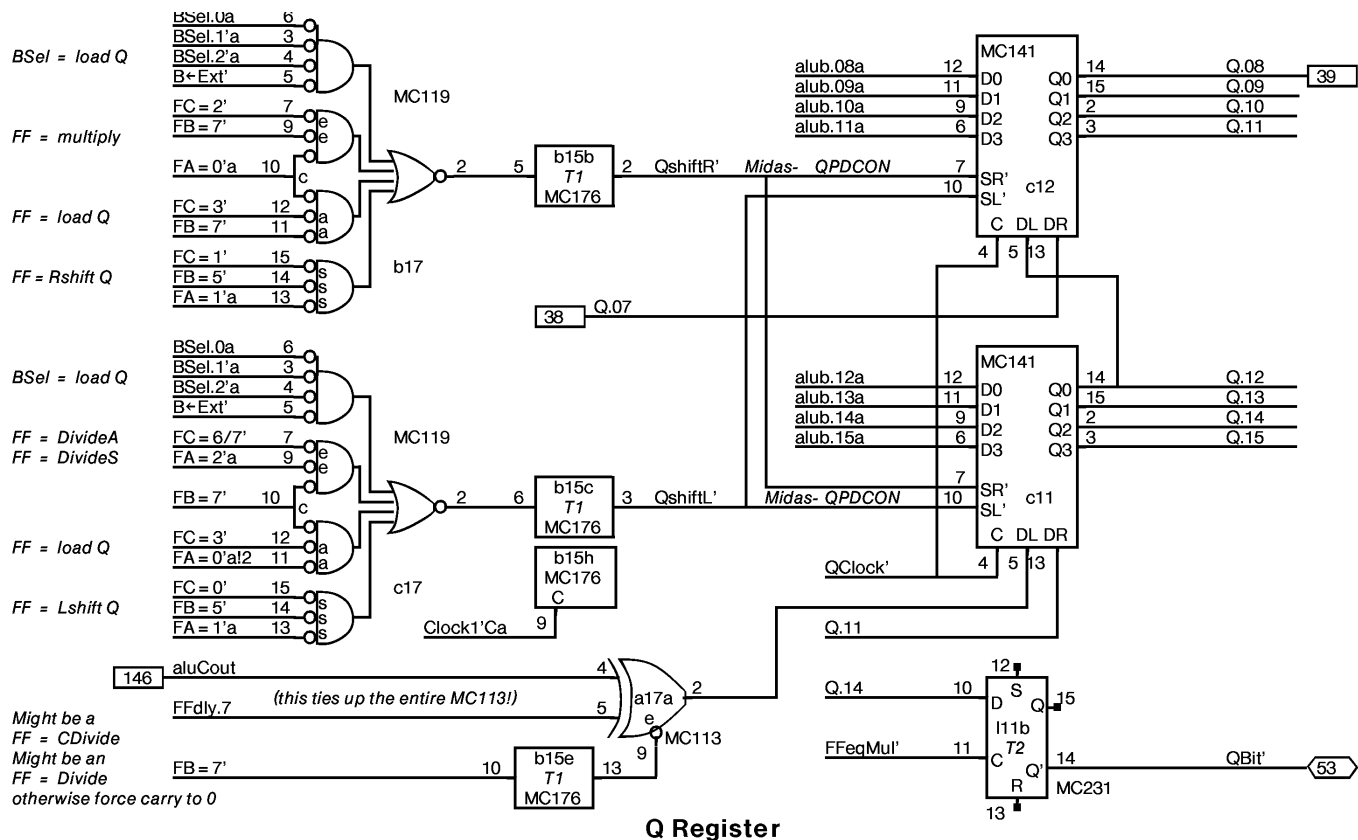
Shc.12-15

	08	09	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
11	1	0	0	0	0	0	0	0
12	1	1	0	0	0	0	0	0
13	1	1	1	0	0	0	0	0
14	1	1	1	1	0	0	0	0
15	1	1	1	1	1	0	0	0
16	1	1	1	1	1	1	0	0
17	1	1	1	1	1	1	1	0
20- 37	1	1	1	1	1	1	1	1

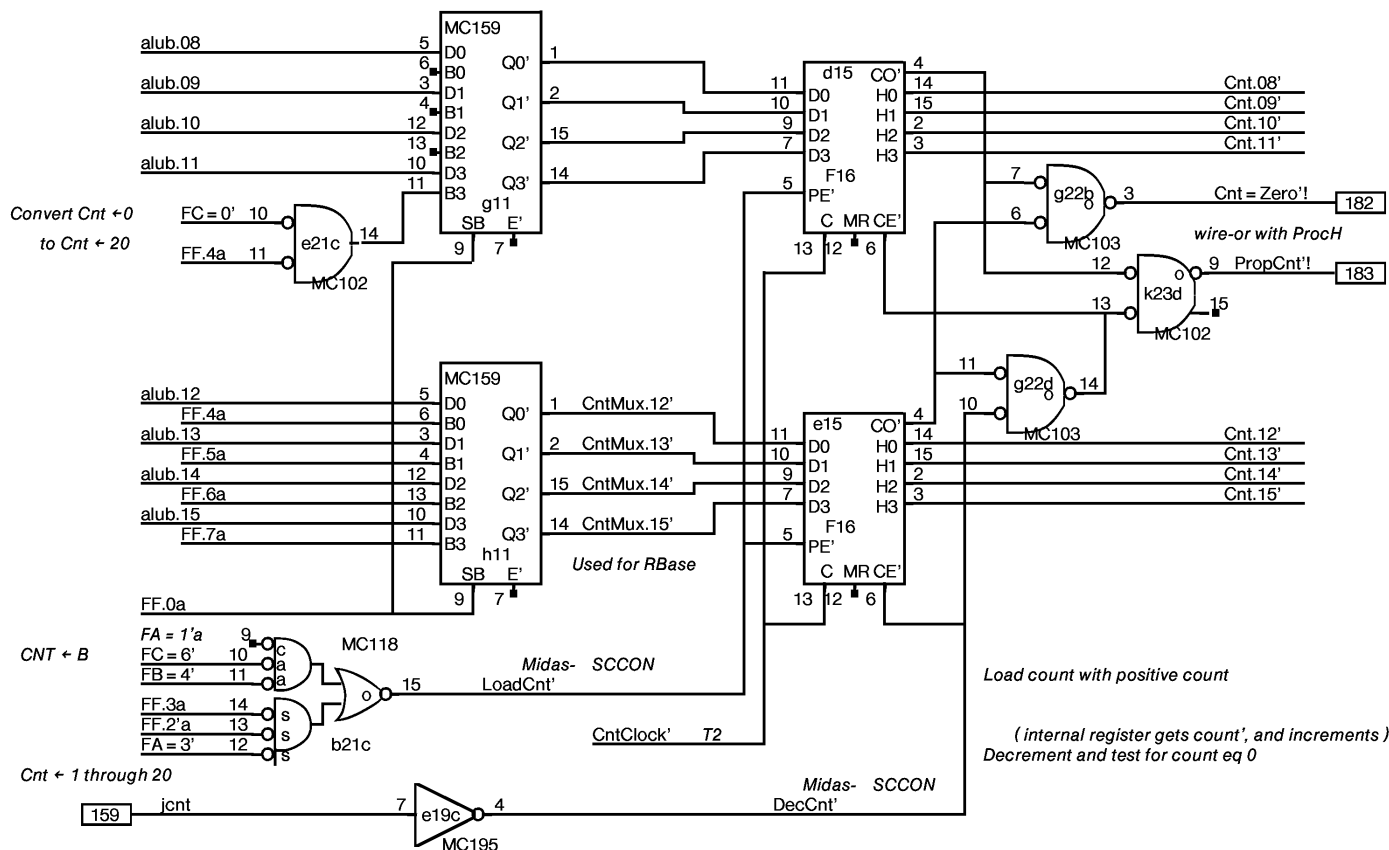
Rmask

Shc.08-11

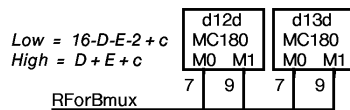
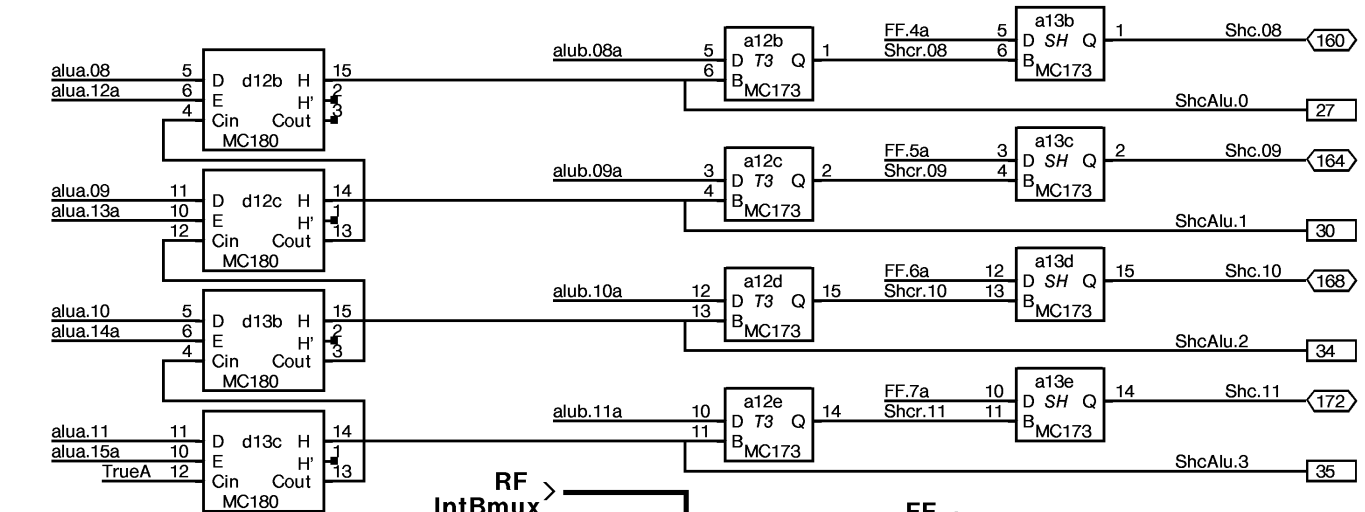
	08	09	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	0	1
3	0	0	0	0	0	1	1	1
4	0	0	0	0	1	1	1	1
5	0	0	0	1	1	1	1	1
6	0	0	1	1	1	1	1	1
7	0	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1
12	1	1	1	1	1	1	1	1
13	1	1	1	1	1	1	1	1
14	1	1	1	1	1	1	1	1
15	1	1	1	1	1	1	1	1
16	1	1	1	1	1	1	1	1
17	1	1	1	1	1	1	1	1
20- 37	1	1	1	1	1	1	1	1



Count Register



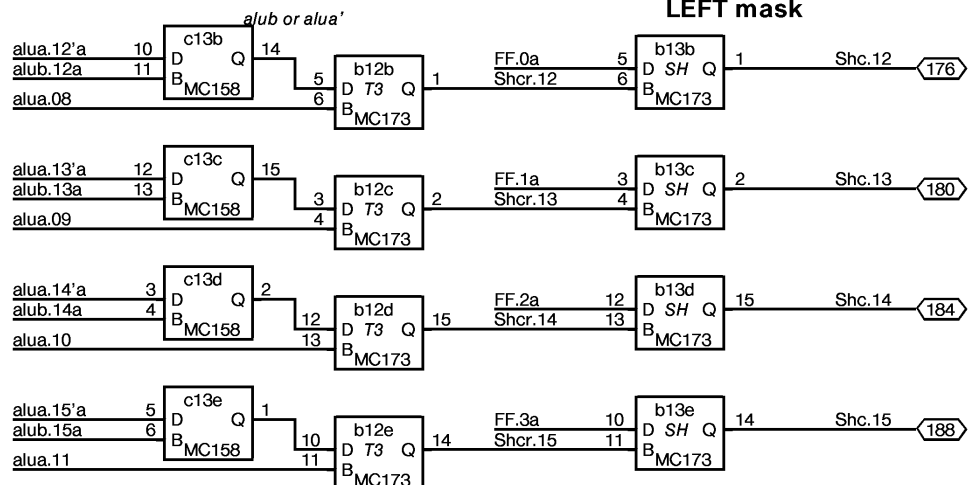
RIGHT mask



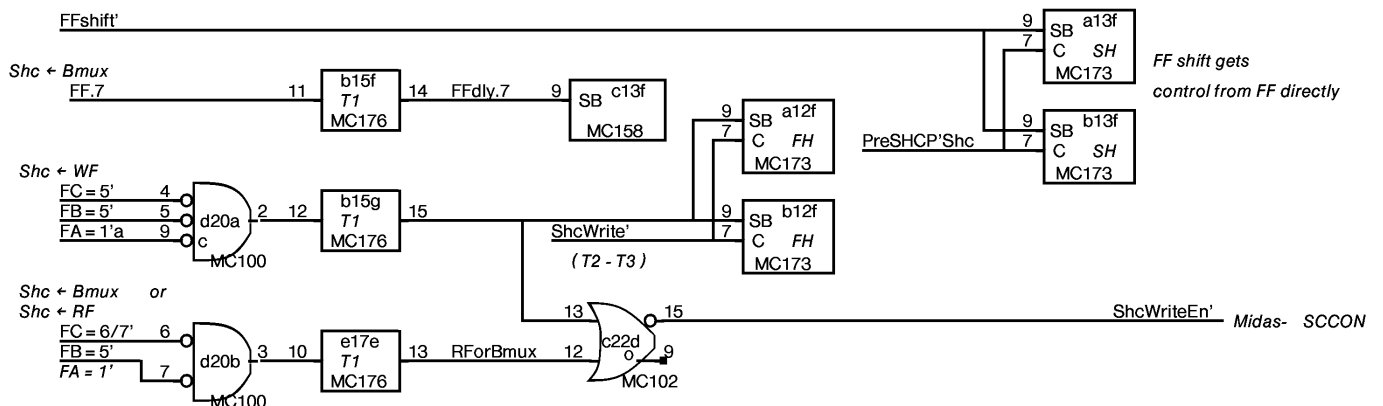
These alu's perform

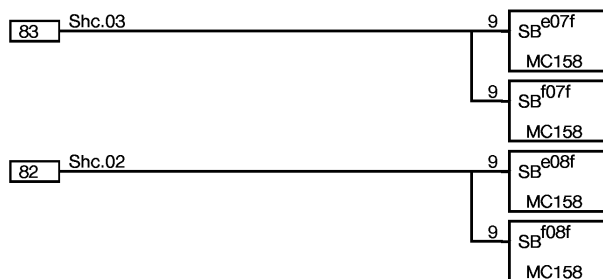
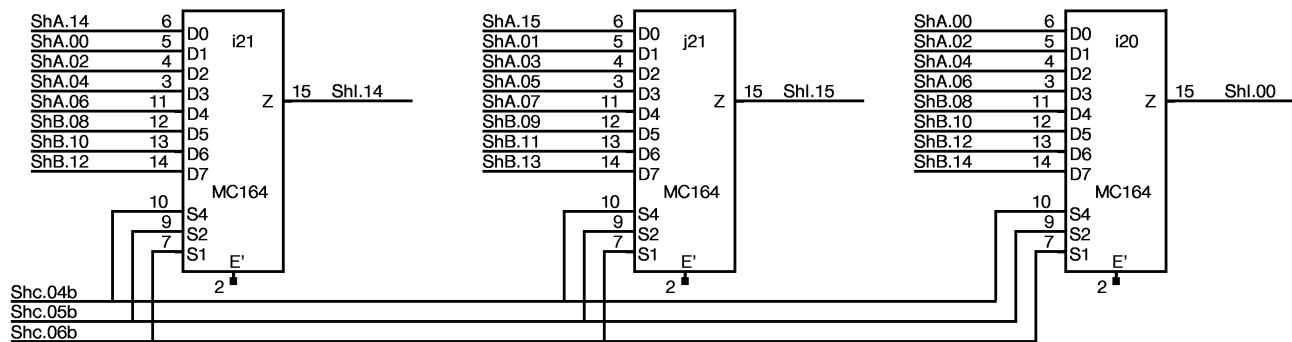
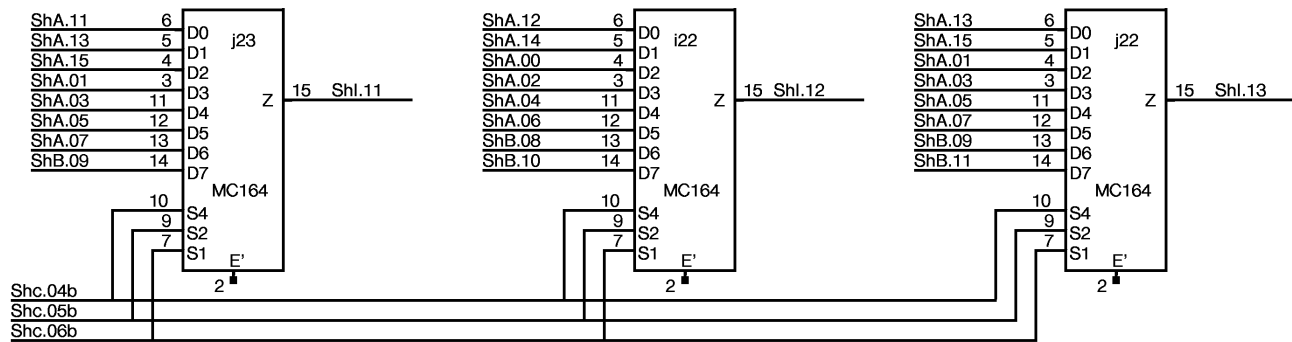
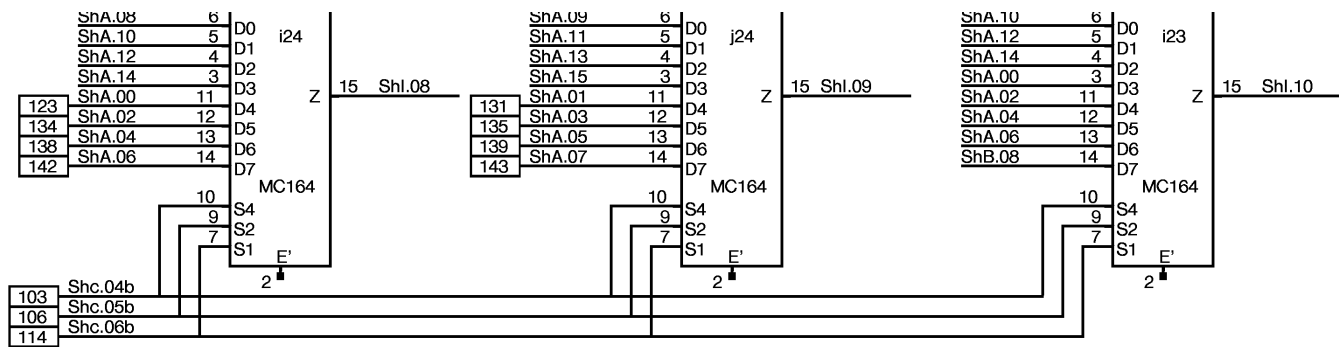
	RF FF = 156	WF FF = 155
Shift Count	$P + S + 1$	$16 - P - S - 1$
Right Mask	Don't Care	$16 - P - S - 1$
LeftMask	$16 - S - 1$ (ie S')	P

"P" = 8:11
"S" = 12:15



Done this way so EF/IF shifts
do not use Shc - IO tasks can
use the shifter without saving
and restoring Shc.

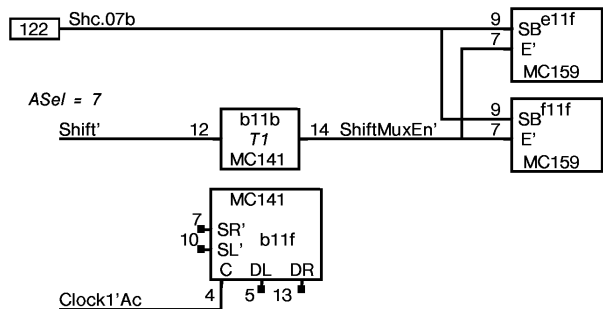




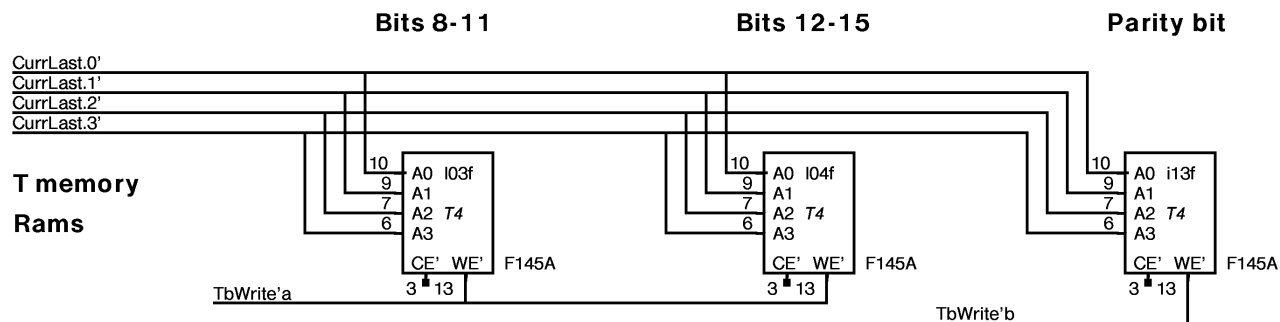
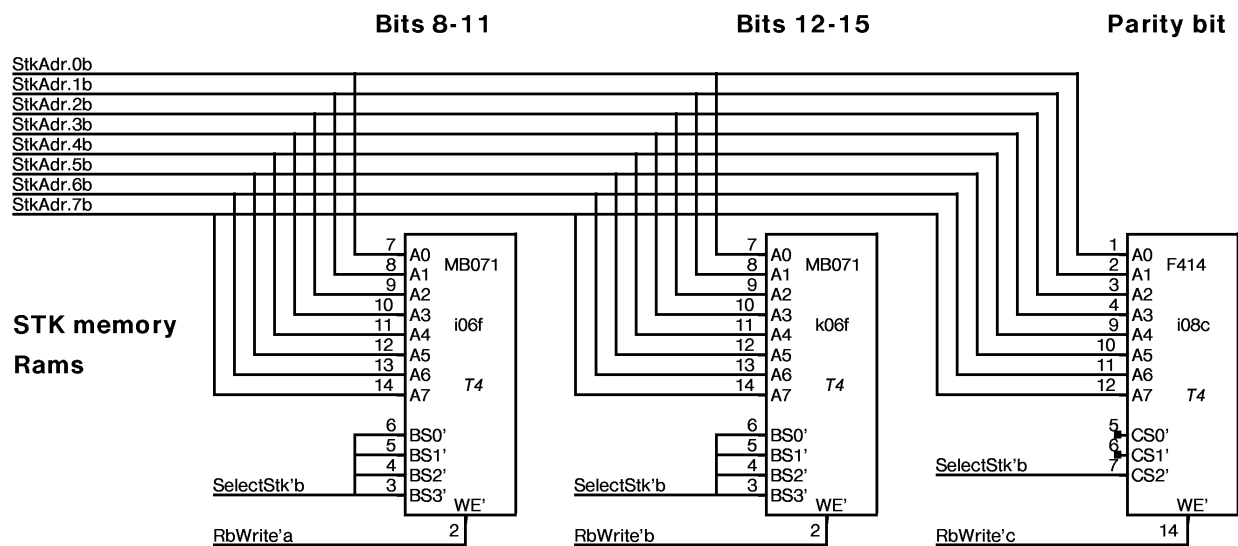
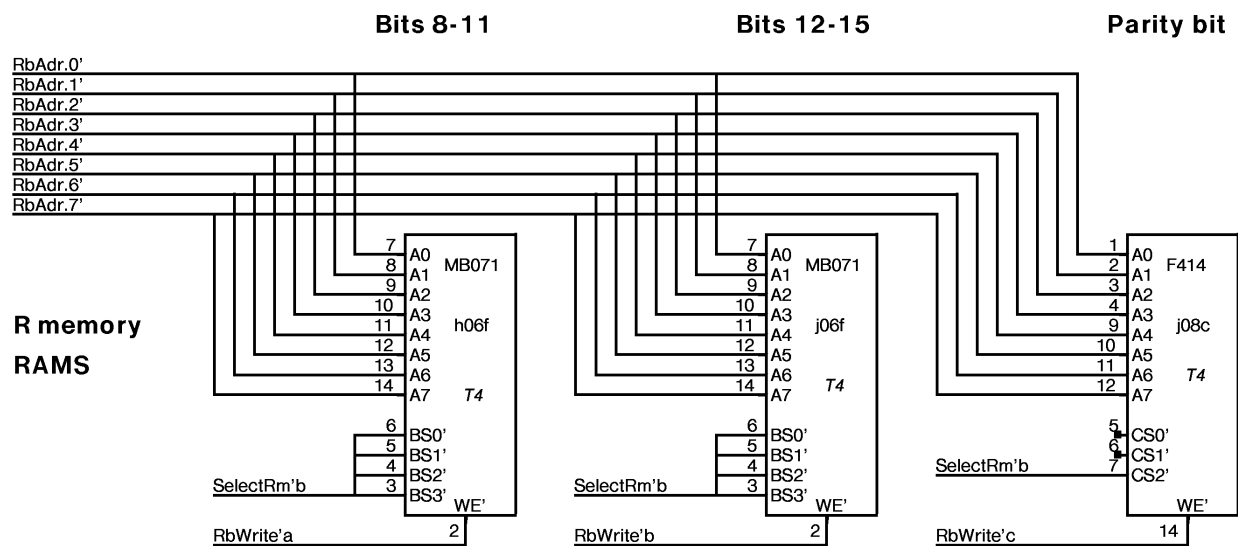
ShA
Mux Control

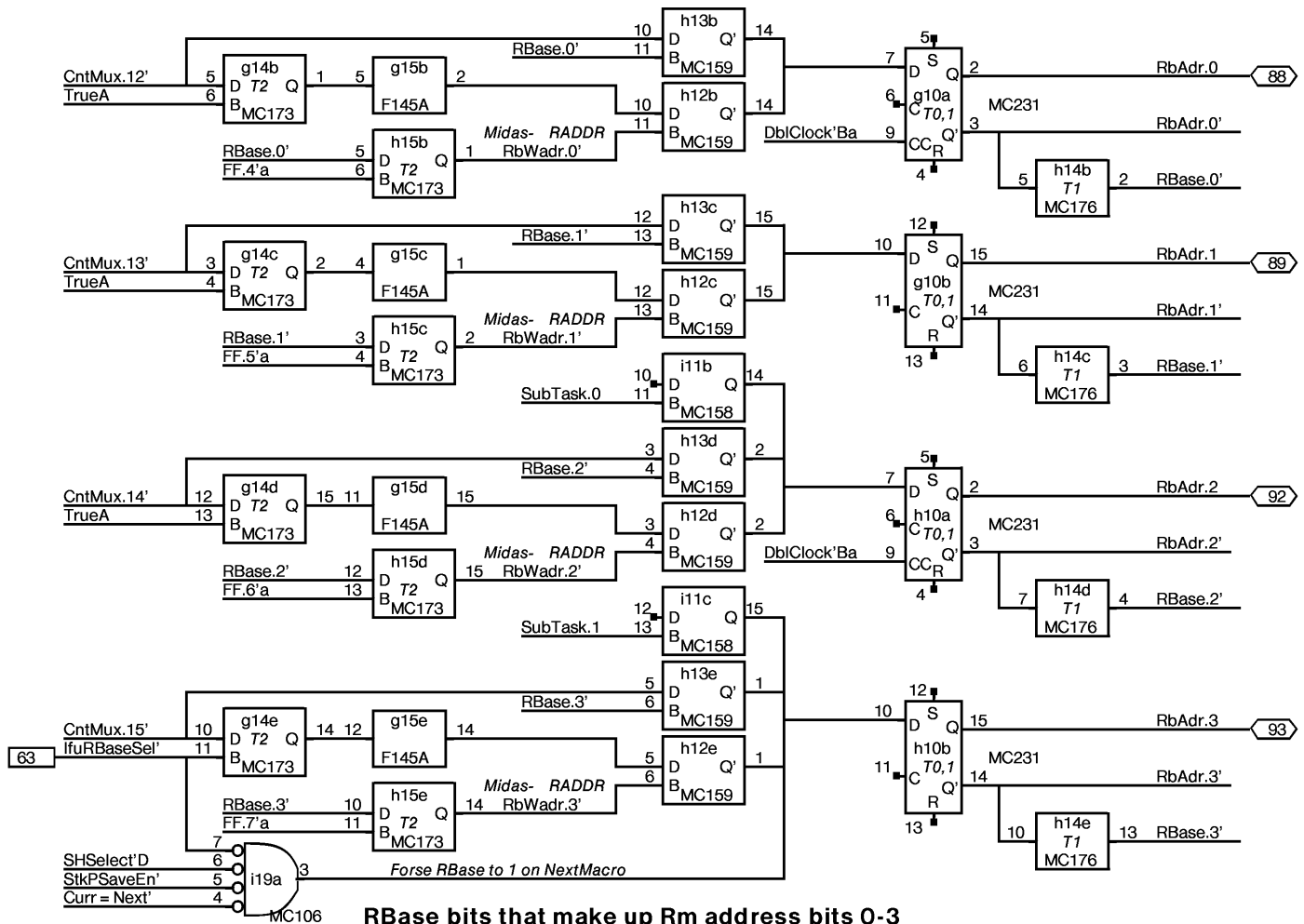
ShB
Mux Control

Shc.02	Shc.03	Shc.04 - .07
0	0	16 bit Left cycle of R
0	1	16 bit Left cycle of R,,T
1	0	16 bit Left cycle of T,,R
1	1	16 bit Left cycle of T

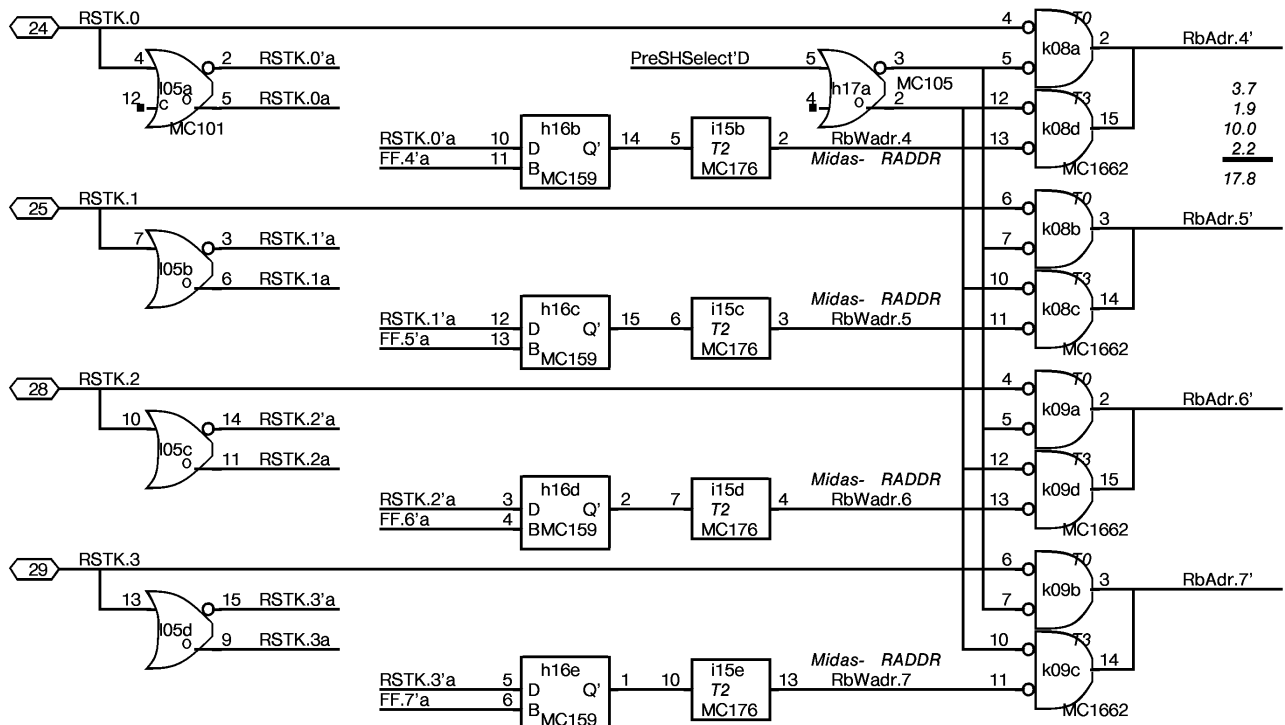


Sh on alua
Mux Control

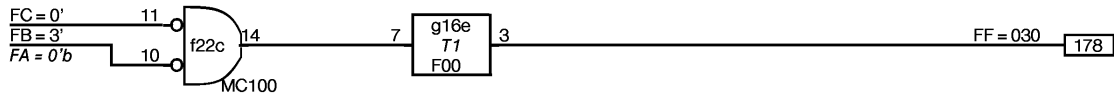




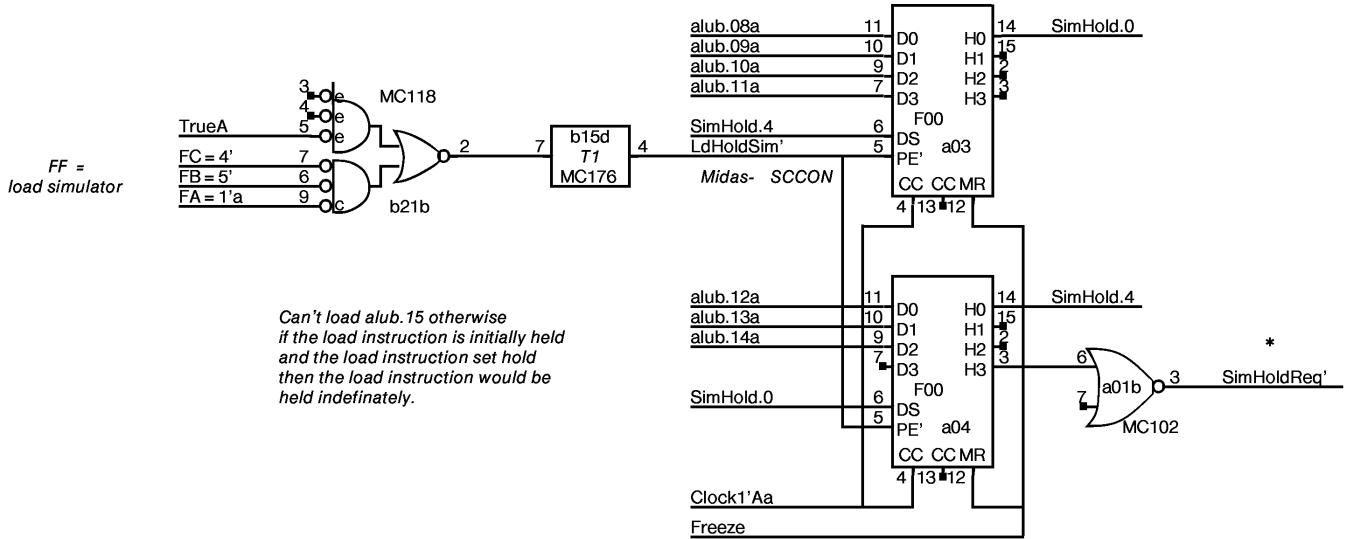
RSTK bits that make up Rm address bits 4-7







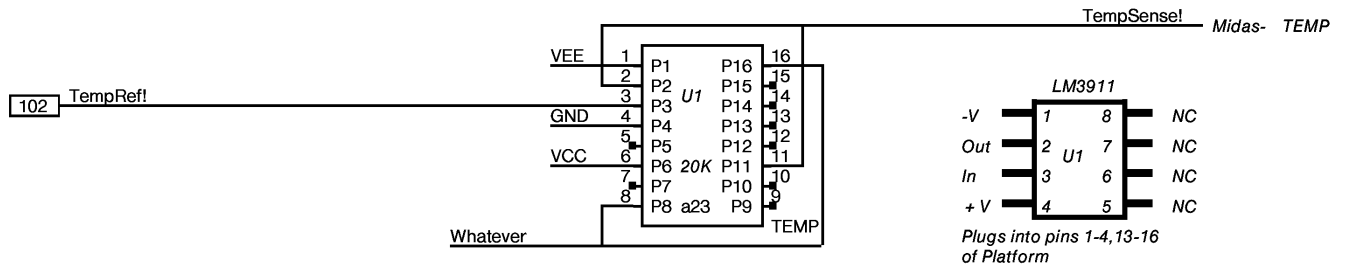
Spare FF Decode

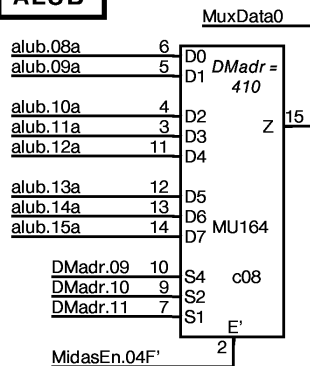
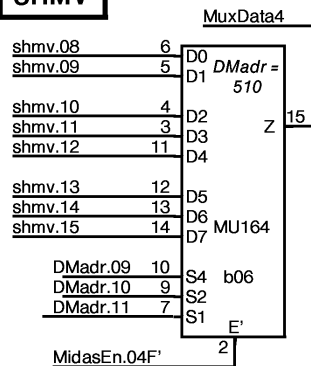
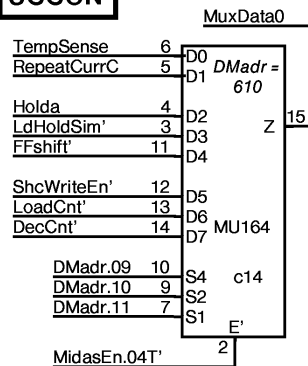
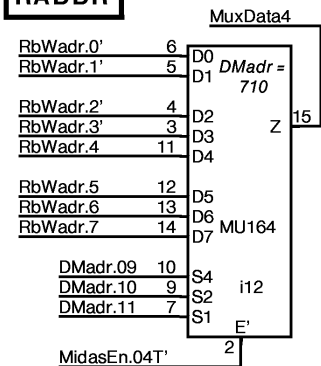
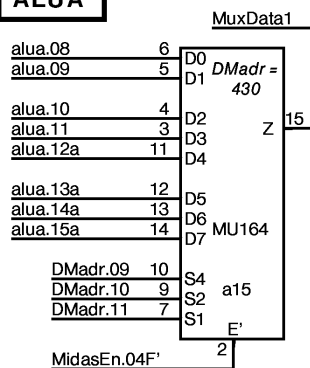
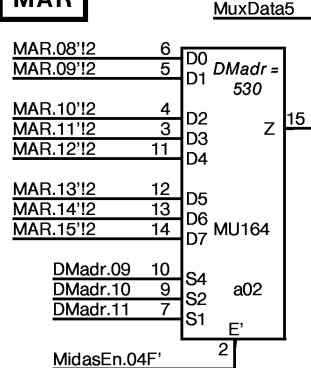
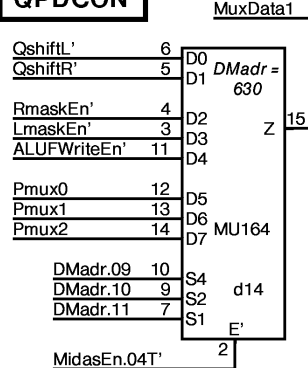
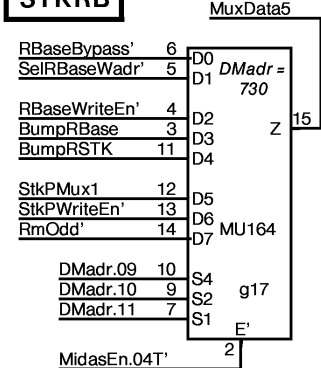
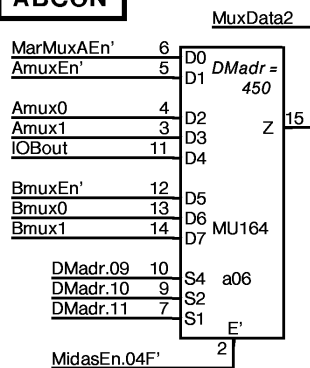
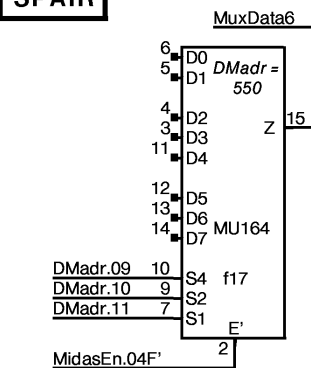
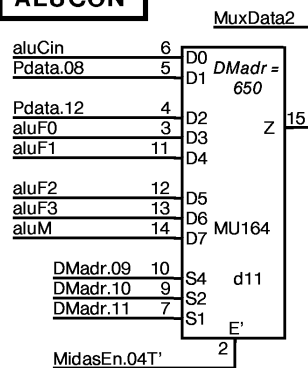
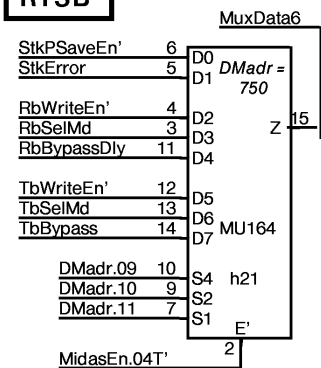
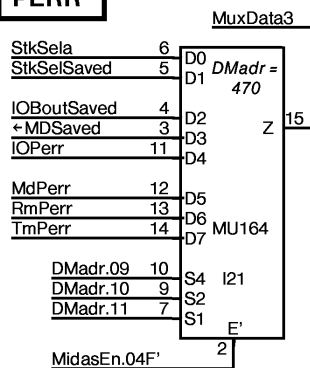
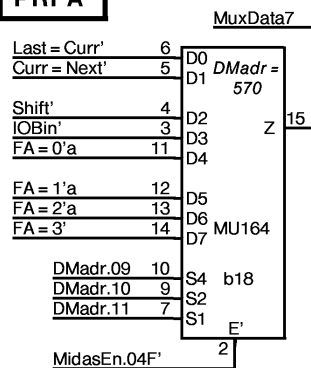
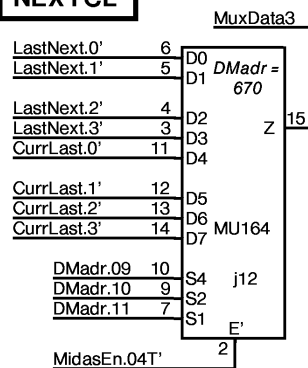
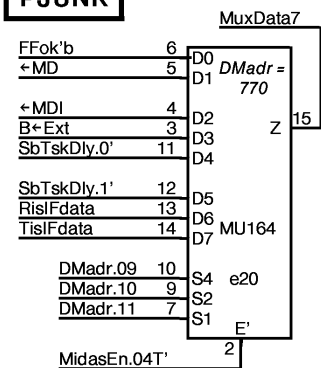


* NOTE: wire [086] with 71 ohms to Gnd
to disable the simulator. (see page 24)

Hold Simulator

Temperature Sensing Ckt



ALUB**SHMV****SCCON****RADDR****ALUA****MAR****QPDCON****STKRB****ABCON****SPAIR****ALUCON****RTSB****PERR****PRFA****NEXTCL****PJUNK**

The Following Signals are provided 50 ohm termination values
by the addition of 100 ohm 1/4 w discrete resistors in parallel with the normal sip

StkSel'a
StkSela
RbAdr.0'
RbAdr.1'
RbAdr.2'
RbAdr.3'
RSTK.0a
RSTK.1a
RSTK.2a
RSTK.3a
RbWadr.0'
RbWadr.1'
RbWadr.2'
RbWadr.3'
RbWadr.4'
RbWadr.5'
RbWadr.6'
RbWadr.7'

For additional information or the rework instructions see
[IVY]<DoradoLogic>ProcL-MWRev-Ch-to-Ci-Rework.sil

-2v TIOA IOB IOB Clk + 12v

	A	B	C	D	E	F	G	H	I	J	K	L	
	184	168	152	136	120	102	84	68	52	36	20		
1	MC102 	Ex Bmux MC174 X	Ex Bmux MC174 X	Ex Bmux MC1662 X	Ex Bmux MC174 X	Ex Bmux MC174 X	IOB par. MC170 X	IOB MC197 X	Md MC175 X	dRm Reg. MC173 X	dTm Reg. MC173 X	Clocks SE210 	1
41	MAR MU164 X	Mar Mux MC159 X	Mar Mux MC159 X	Ex Bmux MC1662 X	Mar Mux MC159 X	Mar Mux MC159 X	Pdata MC164 X	IOB MC197 X	Md MC175 X	dRm Reg. MC173 X	dTm Reg. MC173 X	Clocks SE231 	41
2	SimTW F00 X	Amux MC174 X	Amux MC174 X	Amux MC174 X	Amux MC174 X	Pdata MC164 X	Pdata MC164 X	Pdata MC164 X	T Reg. MC173 X	dT Reg. MC173 X	dRm Par. MC170 X	T Mem. MC145 X	2
42	SimTW F00 X	Bmux MC174 X	Bmux MC174 X	Bmux MC174 X	Bmux MC174 X	Pdata MC164 X	Pdata MC164 X	Pdata MC164 X	T Reg. MC173 X	dT Reg. MC173 X	dT par. MC170 X	T Mem. MC145 X	42
4	Md Parity MC170 X	Amux In MC173 X	Bmux In MC173 X	Amux In MC173 X	Bmux In MC173 X	Pdata MC164 X	dR Reg. MC173 X	dR Reg. MC173 X	Clocks SE212 	Clocks SE210 		RSTK MC101 	4
43	Mux Cont MU164 X	Shmv MU164 X	R Parity MC170 X	T Parity MC170 X	Pdata in MC173 X	Pdata in MC173 X	R Reg. MC173 X	Rm Mem. MB071 X	Stk Mem. MB071 X	Rm Mem. MB071 X	Stk Mem. MB071 X	Rodd MC1668 	43
6		Shmv MC139 X	alub Par MC170 X	alua-a MC101 X	ShA MC158 X	ShA MC158 X	R Reg. MC173 X					Rodd MC1668 	6
7	Clocks SE210 	Shmv MC139 X	alub MU164 X	alu = 0 MC109 	AhB MC158 X	AhB MC158 X	StkPdr T01 MC176 X		Stk parity F414 X	Rm parity F414 X	RbAdr MC1662 X	Rodd MC211 	7
44	Clocks SE210 	alub-a MC1664 X	alub-a MC1664 X	alu	MC181	alu	R/stk sel MC117 X				RbAdr MC1662 X	Rodd MC211 	44
8	Amux T1 MC231 	Bmux T1 MC231 	Mux T1 MC176 				RbAdr MC231 X	RbAdr MC231 X	RBase MC102 		Clocks SE210 	RbBypass MC1660 	8
9	Amux T1 MC231 	Mux T1 MC141 	Q Reg. MC141 X	ALUFM MU164 X	alua sh MC159 X	alua sh MC159 X	CntMux MC159 X	CntMux MC159 X	RBase MC158 X	StkAdr T01 MC176 	StkAdr T01 MC176 	Misc. MC231 	9
45	Shc MC173 X	Shc MC173 X	Q Reg. MC141 X	Shc MC180 X	ALUFM MC176 	Clocks SE212 	Clocks SE211 	RBase MC159 X	RmWadr MU164 X	NextLast MU164 X	RmBypass MC113 X	StkBypass MC113 X	45
10	Shc MC173 X	Shc MC173 X	Shc MC158 X	Shc MC180 X	ALUFM MC145 X	Clocks SE210 	Clocks SE210 	RBase MC159 X	T mem-P MC145 X	StkAdr MC174 X	RmBypass MC113 X	StkBypass MC113 X	10
11	Bmux In MC158 X	Bmux In MC158 X	Shc MU164 X	P mux MU164 X	ALUFM MC145 X	Pdata in MC159 X	RBase MC173 X	RbAdr T1 MC176 	CurrLast' MC141 X	StkAdr MC174 X	StkAdr MC158 X	StkP MC176 X	11
46	alua MU164 X	Q Reg. MC176 		Cnt Reg. F16 X	Cnt Reg. F16 X	Pdata in MC159 X	RBase MC145 X	RBase MC173 X	RBase T0 MC176 	StkAdr MC174 X	Misc. MC107 	StkP MC231 X	46
12						aluCin MC173 	Parity T1 F00 	RBase MC159 X	StkAdr MC173 X	StkAdr MC174 X	StkAdr MC158 X	StkPsave F00 X	12
13	Q Reg. MC113 X	Q Reg. MC119 X	Q Reg. MC119 X	Misc. T0 MC176 	Misc. T1 MC176 , ,	SPAIR MU164 X	Rm cont MU164 X	Misc. MC105 	StkAdr MC173 X	StkP MC182 X	StkP MC182 X	StkP MC182 X	13
47	LastNext' MC158 X	FA MU164 X	Clocks SE210 	Clocks SE210 	P mux MC102 	Misc. MC107 	Misc. T0 MC176 	RBase MC103 	Clocks SE212 	Clocks SE210 	StkP MC195 X	StkPsave F00 X	47
14	LastNext' MC158 X	Misc. MC212 	Amux MC121 X		Misc. MC195 , ,	MC1660	Parity MC117 	Misc. MC102 	Parity MC106 	Ovfl-Ufl MC231 	Ovfl-Ufl MC101 	Ovfl-Ufl MC121 X	14
15	Last' MC141 X	FA = 0 MC100 	FA = 2 MC100 	FA = 1 MC100 	Misc. MU164 X	FA = 0 MC100 	FA = 2 MC100 	FA = 0 MC106 	Shl MC164 X	Ovfl-Ufl MC117 X	Ovfl-Ufl MC119 X		15
16	Curr' MC141 X	FA = 1 MC118 	Amux MC119 X	Amux MC117 X	Misc. MC102 	ALUF MC101 	B+Ext MC109 	Rm cont MU164 X	Shl MC164 X	Shl MC164 X		Parity MU164 X	16
17	Next = Curr MC113 X	P mux MC212 	Misc. MC102 	*MDI MC121 X		FA = 0 MC100 ,	Misc. MC103 	Parity MC231 	Shl MC164 X	Shl MC164 X	Midas MC176 X	Midas MC176 X	17
49	Temp LM3911 X	ASel MC101 	FF dec. MC210 	FF dec. MC101 X	FF-a MC101 X	FF-a MC101 X	Misc. MC102 	LC dec. MC102 	Shl MC164 X	Shl MC164 X	Misc. MC102 	Misc. MC103 	49
18	Next MC101 	ALUF MC211 	Misc. MC103 	FF dec. MC101 X	FF dec. MC161 X	FF dec. MC161 X	MemBase MC231 	Bsel dec. MC101 	Shl MC164 X	Shl MC164 X		Midas MU164 X	18
19													19
50													50
20													20
21													21
51													51
22													22
23													23
52													52
24													24
	10	26	42	58	74	92	110	126	142	158	174		

-5v Muffler + 5v

Spare = 19

XEROX PARC	Project Dorado	Reference Board Layout	File ProcL32.sil	Designer R. Bates	Rev Ci	Date 3/18/80	Page 32
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Use Dorado Proms to define the following Proms:

Board Name	Prom Name	location
PorcL	Lmask (low bite)	b07
	Rmask (low bite)	b08